Canonical bases and geometry

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Lecture 1

Algebraic geometry $\longleftrightarrow$ Representation theory

$\mathfrak{g}$ simple Lie algebra $/\mathbb{C}$

$$\begin{array}{ccc} \cup & & \\ \mathcal{N} & \text{nilp. cone} & \longleftarrow \quad T^*(G/B) \\ \| & & \| \\ Y & \longleftarrow & X \end{array}$$

rep. of $\mathfrak{g}$ = rep of $U\mathfrak{g}$

$U\mathfrak{g}/$ central char. = quantisation of Y

Eg. $\mathfrak{g} = \mathfrak{sl}_2$, $\quad Y = \{xy = z^2\}$

$$U\mathfrak{g}/\text{central char.} = \langle e, f, h \rangle \Big/ \left(\begin{array}{l} [h,e] = 2e, \ [h,f] = -2f \\ [e,f] = h, \ ef + fe + \frac{h^2}{2} = c \end{array} \right)$$

Cat. $\mathcal{O}$ $\supset$ $M_\lambda = \mathrm{ind}_b^{\mathfrak{g}}(\lambda) \longrightarrow L_\lambda$

(principal block of)

$\lambda = w(\rho) - \rho$

Connection to geometry

$$\begin{array}{ccccc} & & & & H^*(X) \\ & & & & \| \\ \# \text{ irr. objects} & = & |W| & = & \dim H^*(G/B) \\ \| & & & & \\ \dim K_0(\mathcal{O}) & & & & \end{array}$$

KL conjectures.

$$K^0(\mathcal{O}) \simeq \mathbb{Z}[W]$$
$$\rotatebox{90}{$\in$}$$
$$[M_{w(\rho)-\rho}] \longleftarrow [w]$$

KL conj. (Th. Beilinson - Bernstein, Brylinski - Kashiwara)

Then $[L_{w(\rho)-\rho}] \longrightarrow$ <u>canonical basis</u>

To define "canonical" need more structure on K^0.

Have an action of <u>braid group</u> B on $D^b(\mathcal{O})$, also on $D^b(U\mathfrak{g}\text{-mod})$
$\qquad\qquad\qquad

Similar story $B \rightsquigarrow B_{aff}$. (before: walls $= x_i = x_j$

$\qquad\qquad\qquad

(Bondal–Orlov, Kawamata)

$$\begin{tikzcd} X \arrow[dr] & & X' \arrow[dl] \\ & Y & \end{tikzcd}$$

two resol'ns of sing. alg. variety Y s.t. $K_X \simeq \mathcal{O}_X$, $K_{X'} \simeq \mathcal{O}_{X'}$.

Then we should have an equiv. $D^b(\mathrm{Coh}(X)) \simeq D^b(\mathrm{Coh}(X'))$.

A more precise conjecture for symplectic resolutions.

X, X' have algebraic symplectic form. Also $X, Y \circlearrowleft \mathbb{C}^*$ so that Y is contracted to a point. $t.w = t^d w$, $d > 0$, $t \in \mathbb{C}^*$ ($d=2$ in example)

(Y normal)

Can define a parameter space S

X, X' $\rightsquigarrow$ (homotopy) point $\in S$ $\ni x$, $\ni x'$

s.t. an equiv. corresponds to a homotopy class of paths from x to x' in S.

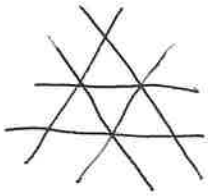
Eg. $X = T^*(G/B)$, $S = \dfrac{\mathfrak{t}^*_{reg,\,aff}}{W_{aff}}$.

Also to other homotopy points, we assign some noncommutative resolutions.

$S \approx$ "Bridgeland" stability space for $D^b(\mathrm{Coh}\, X)$.

More details on construction for $T^*(G/B)$.

a W_{aff} - equivariant local system of categories on $t^*_{reg,aff}$.



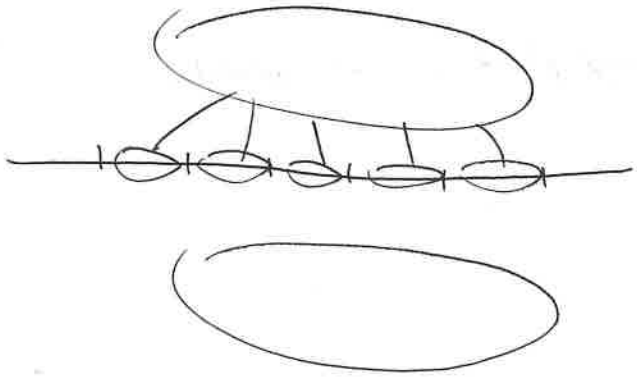
$t^*_{reg,aff} \setminus$ affine coroot hyperplanes

Base points of 2 kinds

1) alcoves - components in $t^*_{reg,\mathbb{R}}$

2) $t^*_{\mathbb{R}} + iC_w$

Weyl cone.



<u>Local system of categories</u>. 2nd kind $\rightsquigarrow$ $(D^b \operatorname{coh} T^*(G/B))$

in general $D^b(\operatorname{coh} X)$.

1st kind $\rightsquigarrow$ NC resolutions

Straight path $\rightarrow$ localization equivalence.

<u>Lecture 2</u>. Picture (Thm for $T^*(G/B)$, almost thm for quiver varieties)

Refined K $\Rightarrow$ D conjecture

$X \longrightarrow Y$ sympl resol'n, $\rightsquigarrow$ local system of categories on a parameter space.

$V = H^2(X, \mathbb{R})$. Have a collection Σ of rat'l hyperplanes H_α, which partition

V into cones, s.t. $C \rightsquigarrow X_C$

$\quad\quad\quad

Recall we construct a local system of categories $(V_{\mathbb{R}} + iC) \rightsquigarrow D^b(\mathrm{Coh}(X_C))$.

alcoves $\rightsquigarrow$ NC resolutions constructed via char p quantizations

Have a family of quantizations $\mathcal{O}^{\lambda}_{\mathrm{quant}}(V) \simeq \Gamma(\mathcal{O}^{\lambda}_{\mathrm{quant}}(X))$

$\lambda \in \Lambda$

For quiver varieties, Losev's talks

For $T^*(G/B)$, $\mathcal{O}^{\lambda}_{\mathrm{quant}}(X) = \mathrm{Diff}_{\lambda}(G/B)$

$\mathcal{O}^{\lambda}_{\mathrm{quant}}(Y) = U_{\lambda}(\mathfrak{g})$

Observation:

$$\begin{tikzcd} X \arrow[dr] & & X' \arrow[dl] \\ & Y & \end{tikzcd} \qquad \begin{array}{ccc} X & & X' \\ \cup & & \cup \\ U & \simeq & U' \end{array}$$

codim 2

$\Downarrow$

quantizations for X, X' are in bijection s.t. global sections are the same.

<u>λ is good</u> (for X) if $R\Gamma : D^b(\mathcal{O}^{\lambda}_{\mathrm{quant}}(X)) \xrightarrow{\sim} D^b(\mathcal{O}^{\lambda}_{\mathrm{quant}}(Y))$

<u>Prop.</u> λ is good iff $\mathcal{O}^{\lambda}_{\mathrm{quant}}(Y)$ has finite global dim. ($\Rightarrow$ indep. of X)

For a good λ, we get

$$\begin{tikzcd} D^b(\mathcal{O}^{\lambda}_{\mathrm{quant}}(X)) \arrow[rr, phantom, "\simeq"] \arrow[dr, "\sim"'] & & D^b(\mathcal{O}^{\lambda}_{\mathrm{quant}}(X')) \arrow[dl, "\sim"] \\ & D^b(\mathcal{O}^{\lambda}_{\mathrm{quant}}(Y)) & \end{tikzcd}$$

Use char. p to pass from $\mathcal{O}^{\mathrm{quant}}(X)$-mod to Coh

p-centre phenomenon (e.g. for $k[x,\partial]/[x,\partial]=1$)
$\boxed{\text{char } k = p > 0}$

<u>Prop</u> 1) $\mathcal{O}^{\lambda}_{\text{quant}}(X)$ is an Azumaya algebra of rank $p^{\dim X}$ on $X^{(1)} = X \times_{\operatorname{Spec} k} \operatorname{Spec} k$, $\operatorname{Spec} k \xrightarrow{Fr}$

2) This AA splits on formal nbhd of every fibre of $\pi: X \to Y$.

Moreover, its class is $Br = \pi^*(C)$, $C \in Br(Y)$

(BMR for $T^*(G/B)$
Kubrak - Traskin for quivers)

(replace $D^b(Coh(X))$ by $D^b(Coh_0(X))$)
supported on $\pi^{-1}(0)$

<u>So</u> for every good λ, get an equiv.

$$D^b\, Coh_0(X) \simeq D^b\, Coh_0(X') \quad \text{from } Coh_0(X) \simeq \mathcal{O}^{\lambda}_{\text{quant}}(X)_0\text{-mod}$$

$\mathcal{F} \longmapsto \mathcal{F} \otimes \mathcal{E}$, $\mathcal{E}$ - splitting bundle.

& same for X'.

<u>Ex.</u> $X = T^*(M)$, $M = G/P$, $\mathcal{E}|_M = Fr_*(\mathcal{O}(\lambda))$

Σ

Which λ's are good?

char $k = p$. goodness depends on λ mod p.

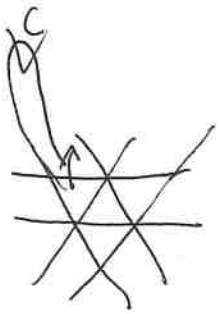
<u>Claim</u> 1) λ is good when $\frac{\lambda}{p}$ is not near a hyperplane $H \in \tilde{\Sigma}$.

2) The equivalence $D^b(Coh(X)) \overset{\varphi_\lambda}{\simeq} D^b(Coh(X'))$

depends only on the alcove of $\frac{\lambda}{p}$.

3) The equivalences lift to char. 0.

Idea of the proof:



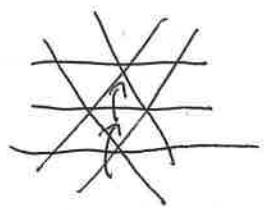
<u>Th</u>. monodromy of QH_T^* $\simeq$ the induced action on K^0

(use shift operator to make sense of it)

Perverse equivalences & central charges

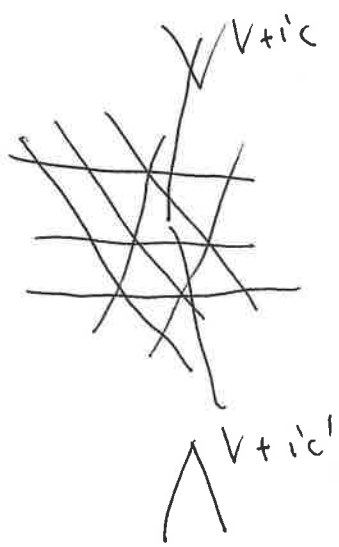
Say more about "half-monodromies"

This is a perverse equivalence, governed by a central charge (real variations of stabilities)



<u>Lecture 3</u>. <u>Th</u>. Monodromy of q. conn. $\simeq$ action of π_1 on $K_T(X)$.

<u>Idea</u>: compute the isomorphism $K_T(X_c)$ & $K_T(X_{c'})$ via an alcove A.



The equivalence $D^b Coh^T(X_c) \xrightarrow[\sim]{\Phi_A} D^b Coh^T(X_{c'})$ categorical

sends stable envelopes to stable envelopes.

for stability parameter $\lambda \in A$.

(assume X^T is finite)

$$D^b\,\mathrm{Coh}(X_c) \xrightarrow[\sim]{\Psi} D^b\,\mathrm{Coh}(\mathcal{A}_A) \xrightarrow[\sim]{\Psi'} D^b\,\mathrm{Coh}(X_{c'})$$

NC-resol'n corr. to A

$\Phi_A = \Psi' \circ \Psi$, $\quad St_x^A \overset{\Psi}{\mapsto} M_x \overset{\Psi'}{\longmapsto} St_x^A$.

Verma module

categorical Stable envelope

characterized by:

$D^b\,\mathrm{Coh}^T(X) \ni St_x^A$ supported on $\bigcup_{y \le x} \mathrm{Attr}_y$ — closed Lagrangian

St_x^A near x is $\simeq$ an explicit line bundle

$\mathrm{wts}\left(St_x^A\big|_y\right) \subset P_y + \mathrm{Shift}(\lambda)$

$y \le x$ $\qquad$ polytope, $\lambda \in A$

Define shift for $\lambda \in \Lambda = \mathrm{Pic} \subset V$

$\mathrm{Shift} = \mathrm{wt}\left(\mathcal{O}(\lambda)_y\right) - \mathrm{wt}\left(\mathcal{O}(\lambda)_x\right)$.

[Halpern-Leistner, Mauldin, Okounkov]

Central charges & <u>perverse equivalences</u>

Recall that for abelian categories A, B, $D^b(A) \overset{\Phi}{\simeq} D^b(B)$ is a perverse equiv. corr. to filtrations A_i, B_i. if $gr_i(\Phi)[i]$ is t-exact.

Let $\mathcal{A}$ be an abelian cat.

$$\zeta : \mathbb{R} \longrightarrow K^0(\mathcal{A})^*_{\mathbb{R}} \quad \text{s.t.}$$

$$\langle \zeta(t), [M] \rangle > 0 \quad \text{for } 0 \neq M \in \mathcal{A}, \quad t \in (0, \varepsilon). \text{ some } \varepsilon.$$

Then let $\mathcal{A}_i = \{ M : t \mapsto \langle \zeta(t), [M] \rangle \text{ is } O(t^i) \}$

Define $Z : \overset{H^2(X;\mathbb{R})}{\overset{\shortparallel}{V}} \longrightarrow K^0(\mathrm{coh}_0(X))^*_{\mathbb{R}}$ by

$$\lambda \longmapsto \Big[[F] \mapsto \underset{\substack{\uparrow \\ \text{Euler char.}}}{\chi}(F \otimes \mathcal{O}(\lambda)) \Big]$$

<u>Claim</u> (Th. for $T^*(G/B)$)

[BL 2017]

ζ is a perverse equivalence, for the filtration as above.

$\zeta = Z|$ line through the face.

<u>Rmk.</u> Relation to Rep. Theory.

$$\mathcal{A}_A \simeq \mathcal{O}^{\lambda}_{\gamma, \text{quantum}-\text{mod}_0}, \quad \lambda \in \rho A$$

$$\Downarrow$$

$$M$$

$$d_M(\lambda, \rho) = \dim M_{\lambda, \rho}$$

$$\Big|$$

polynomial.

Want to count irreducibles L for a given deg$_p$ $d_L(\lambda, p)$.

deg $= 0$ $\iff$ L comes from char. 0

deg $= a$ $\longleftrightarrow$ # of objects in cat. $\mathcal{O}$ over $\mathbb{C}$ w/ supp. of dim a.

2) rel'n to Bridgeland stability.

Real variation of stabilities [ABM]

t-structure. $\zeta \in (\mathfrak{t}^\circ)^*$

<u>Rmk 3</u> rel'n to connection

Z is a flat section of a ~~trivial~~ specialisation of the connection.

Want to check that the basis of irreducibles in $\mathcal{A}_A$ is a canonical basis.

in $K_T(\mathcal{A}_A) \simeq K_T(X_{C'})$

<u>Th</u> [B.-Mirković] for $T^*(G/B)$

In that case the thm is reduced to

<u>Conj.</u> $\mathcal{A}_A$ is Koszul $\iff$ positively graded

$\mathcal{A}_A = R_A\text{-mod}$, $R_A = \mathrm{End}^{op}(\mathcal{E}_A)$

vector bundle on X_C

Consider $\mathbb{C}^* \curvearrowright X, Y$ dilating the form. $\mathcal{E}_A$ can be given a $\mathbb{C}^*$-equiv. str.

R_A graded.

R_A gets a grading.

Conj. says this grading can be chosen s.t. $(R_A)_{<0} = 0$, $(R_A)_0$ semisimple

Then R_A is Koszul quadratic algebra.

Combinatorics for $X = \mathrm{Hilb}^n(\mathbb{A}^2)$ Dumanski, Yue